GREY'S ALGORITHM

A FASTER WAY TO IDENTIFY BREAST CANCER

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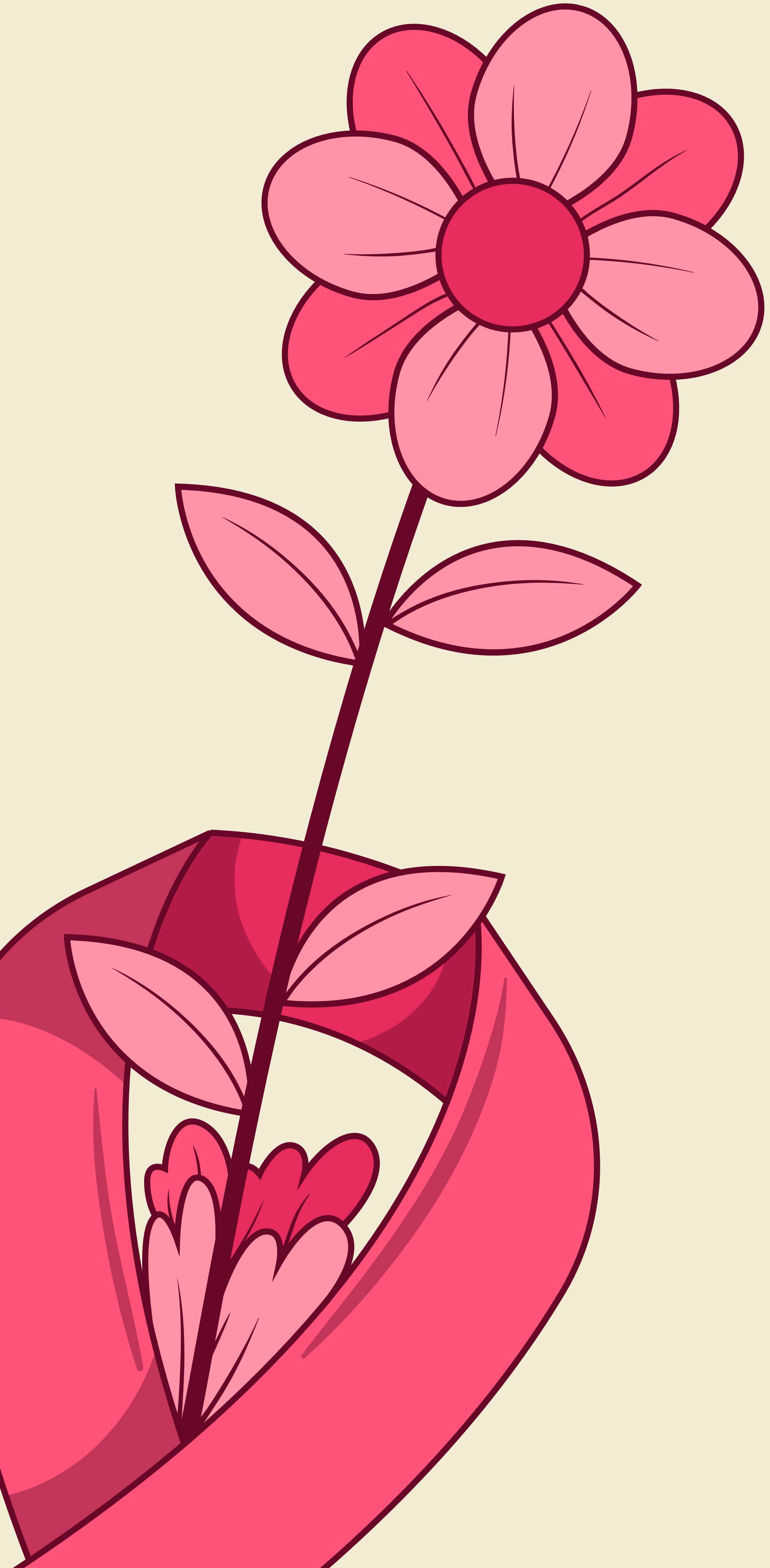
THE CHALLENGE

Breast cancer diagnosis today

- Breast cancer is one of the **most common** cancers worldwide
- Early detection saves lives - **early-stage survival ~99%**
- Radiologists face **heavy workloads** and **time pressure**
- There is a **shortage of expert radiologists**, especially in low-resource areas or rural clinics
- Subtle tumor margins can be **missed** by the human eye

OUR SOLUTION



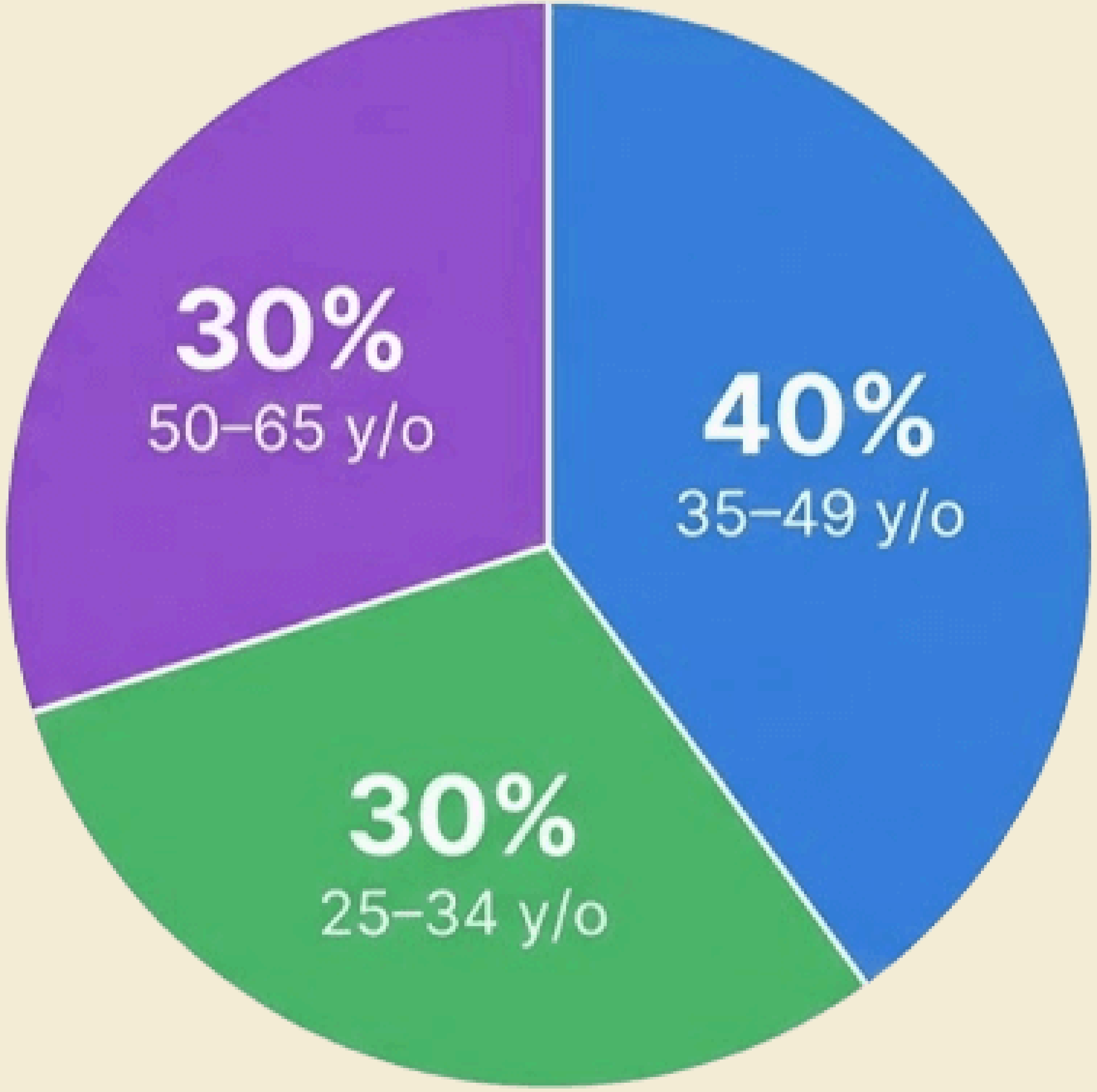
- **Detection**: Our AI-powered application enables *earlier diagnosis* through *automated tumor and calcification segmentation*
- **Reducing radiologists' workload**: It provides a *fast second opinion* and is *easy to integrate* into existing workflows
- **Clear, explainable visual results**: It addresses *clinician fatigue*, helps with *ambiguous cases,* and *eases patient anxiety* caused by long waiting times
- **Accessible anywhere**: This easily deployable tool will help clinics move beyond manual-only diagnosis

Market Description - Romania

- **6.000 - 12.000 new cases** anually
- **3.900** anual deaths (2020)
- **425** hospitals, of which only **60+** can treat/diagnose breast cancer

Demographic Segmentation

Age Distribution



Competitors

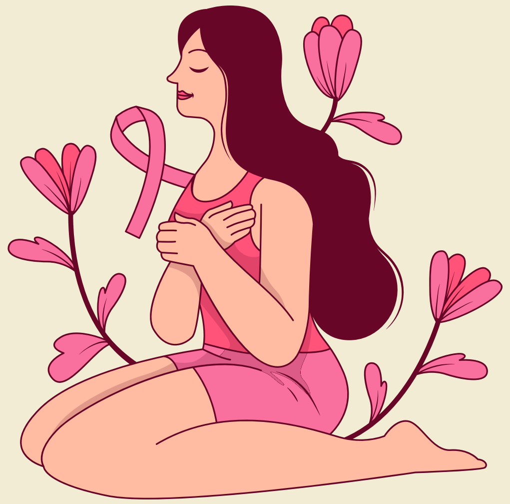
Population Health

SmartMammo™

- High Performance
- Cloud architecture implementation
- Specializes in dense breast tissue
- Lacks an anomaly classification function
- Uses bounding boxes instead of a more precise form of segmentation

RAYSCAPE

- Industry recognition and clinical evidence
- Robust integration of visualisation tools, usable for a more intricate exploration of the area.
- Only available for lung and chest ex-rays, and not specialized for mammograms



Our solution

- Transparency present every step of the way
- Specialized in tumor segmentation, using pixel level annotations
- Clear visuals and easy to navigate

BUSINESS MODEL

***Value Proposition*: Affordable, AI-powered breast cancer detection that provides fast, explainable second opinions and reduces radiologists' workload - deployable in any clinic**

Key Partners: Hospitals and imaging clinics

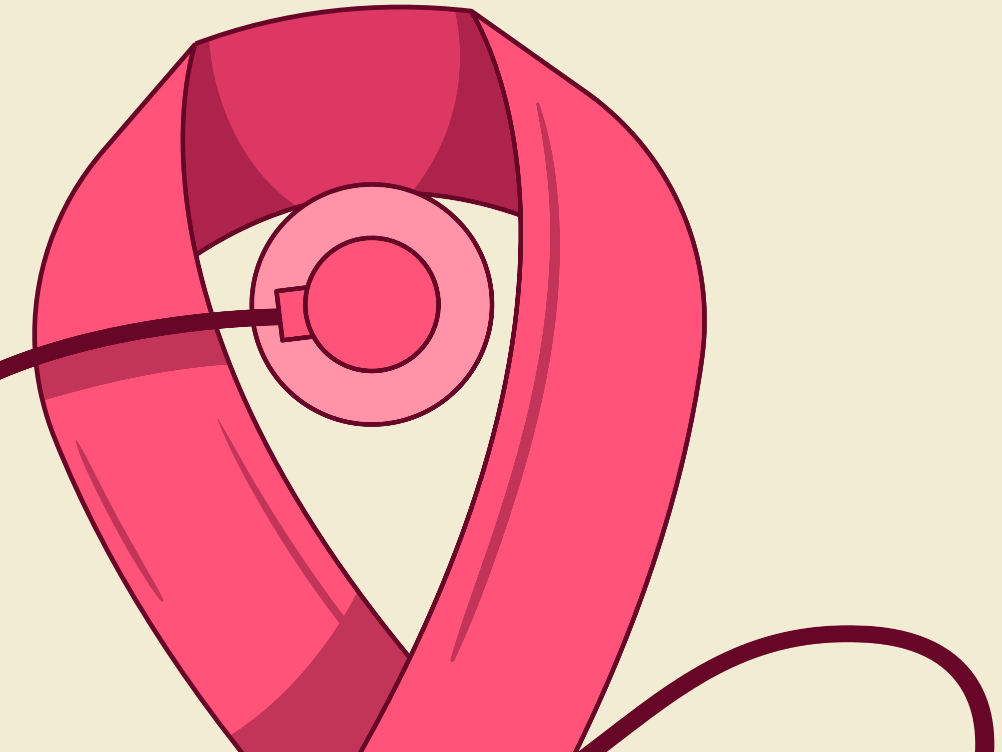
Key Resources: Adaptable AI models, cost-optimized cloud infrastructure

Key Activities: Integrating AI into clinical workflows, providing performance and accuracy statistics, ensuring data security and patient privacy

Customer Relationships: Transparent communication and pricing, long-term partnership mindset

Channels to Customers: Direct sales to clinics and hospitals, digital marketing

Market Segments: Small and medium imaging clinics, public hospitals, rural and low-resource healthcare providers



Monetization Model

- Monthly subscriptions (scan-based tiers)
    a. Basic Plan: 300 scans/month
    b. Growth Plan: 800 scans/month
- Pay-per-extra scan usage
- Research and institutional contracts

VGG16 is our base, a deep learning model originally trained on the ImageNet dataset. This model provides our segmentation for mammography after being fine-tuned using the **InBreast** and **DMID** datasets.

The model uses a hybrid architecture:

- Encoder (VGG16) - to identify formations and texture in the image
- Decoder (U-Net) - to draw the box around the suspected area
- Headless Integration - We "tricked" the VGG16 classification model by removing its top dense layers, repurposing it to output spatial feature maps instead of class labels.

HOW DOES OUR MODEL WORK?

Image Preprocessing

Original

Flipped + Resized

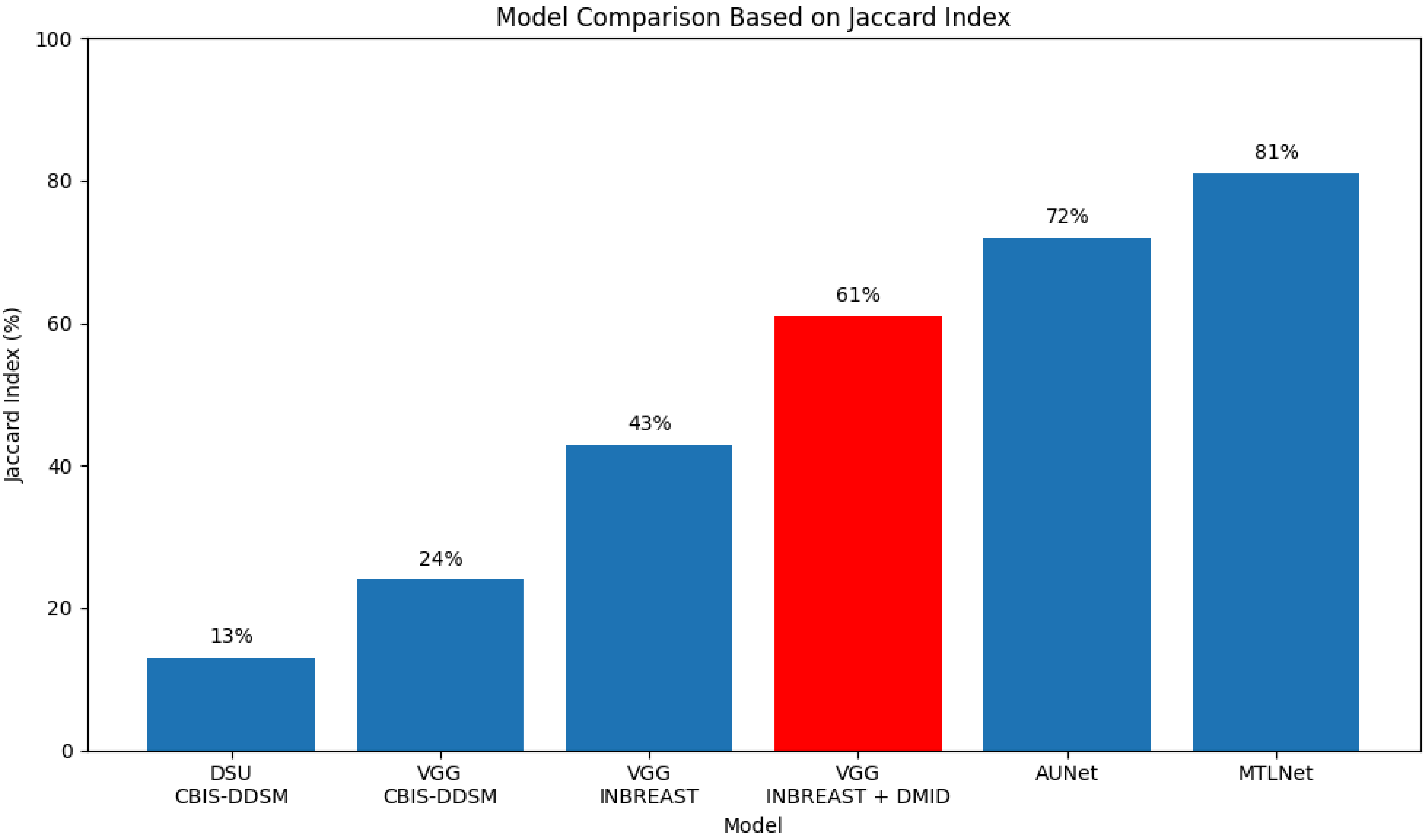
Artifact mask applied

CLAHE filter

HOW DOES OUR MODEL WORK?

Measurement functions to identify not only if the model found the tumor, but also how precisely it covered it:

- Dice Loss: the error function - it checks whether the model gave the coverage as similar as the doctor. The smaller the value of Dice loss, the better the model
- IoU Score (Intersection over Union) -the success metric



Model Comparison Based on Jaccard Index
Jaccard Index (%)
100
80
60
40
20
0
13%
24%
43%
61%
72%
81%
DSU
CBIS-DDSM
VGG
CBIS-DDSM
VGG
INBREAST
VGG
INBREAST + DMID
AUNet
MTLNet
Model

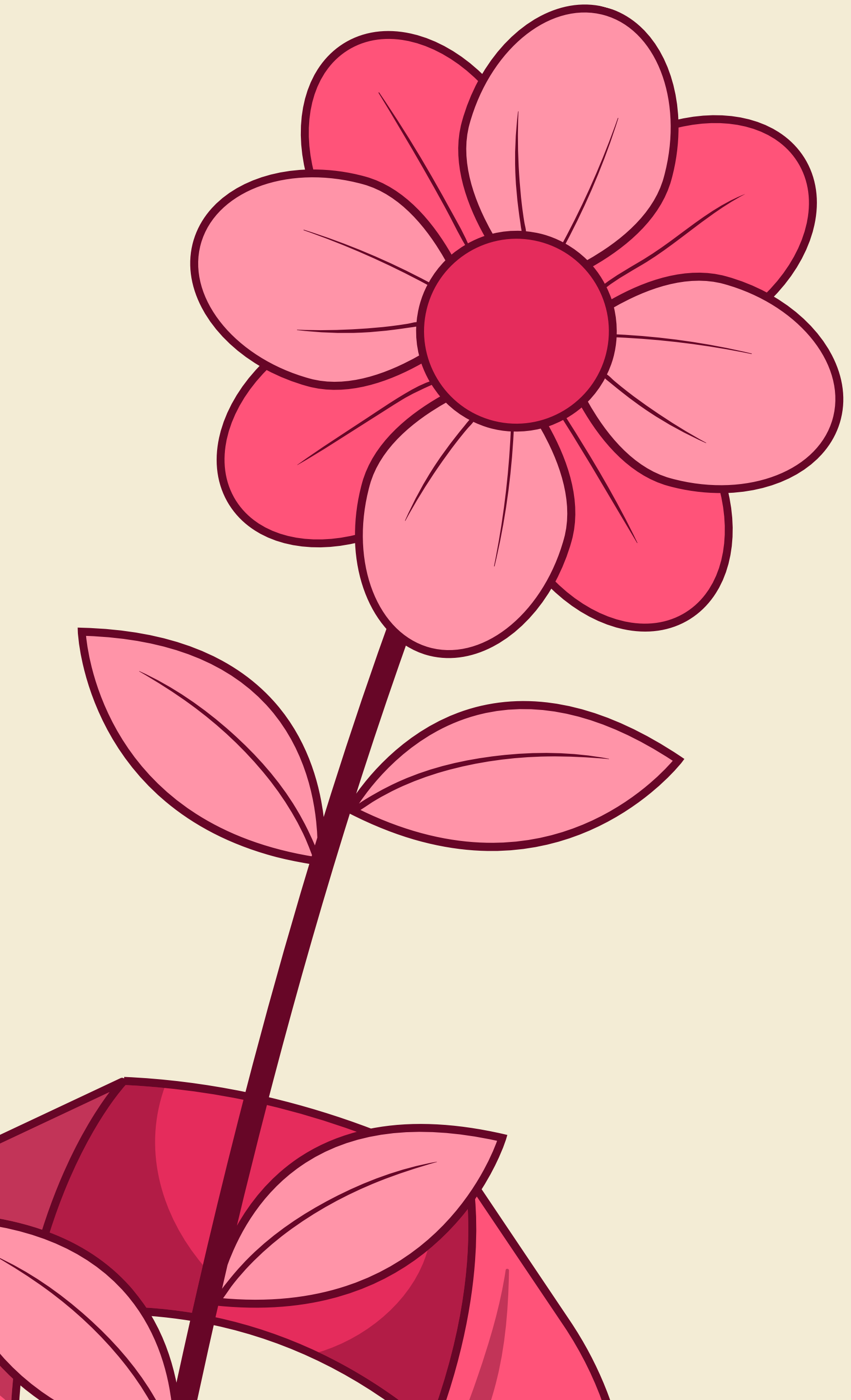
HYPER PARAMETER ANALYSIS

Category	Parameter	Value
Architecture	Backbone	VGG-16 (ImageNet)
	Input Size	512 × 512 × 1
Training	Learning Rate	1×10^{-4}
	Batch Size	4
	Epochs	40 (early stop: 5)
Loss	Function	BCE + Dice
Preprocessing	Clahe	2.0 clip, 8×8 grid
Data Split	Train / Val / Test	80% / 10% / 10%

APP COMPONENTS

- **Minimum Functions**: Scan Management, DICOM visualization, AI-based tumor segmentation with visual overlay
- **Self Made Components**: AI backend, data processing pipeline, WinForms medical viewer
- **Innovation Component**: Highlighting suspicious regions by generating a binary segmentation mask
- **Deployment**: Pyhton server, using a FastAPI AI backend, and a Windows Desktop Client connectet via REST APIs

FUTURE WORK



- **Advanced Deployment:** Implementing a server-side deployment to transform the local framework into a globally accessible application for remote clinical use.
- **Multi-Class Classification:** Expanding the model to distinguish between **Malignant**, **Benign** and **Calcifications** for more granual diagnostic support.
- **Architecture Evolutions:** Integrating the proposed **DSU-Net** to leverage spatial attention gates for precise lesion segmentation.
- **Clinical Robustness:** Increasing dataset diversity and training epochs to improve accuracy and ensure reliability across different mammography scanners.

Bibliography

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